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Paleomagnetism of Paleozoic sedimentary rocks from the Karatau Range, Southern Kazakhstan: Multiple remagnetization events correlate with phases of deformation

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[1] The paleogeography of the Altaids and its kinematic and tectonic evolution during the final collision and amalgamation of Eurasia is still poorly known. Addressing this problem, a paleomagnetic study has been undertaken on Paleozoic sedimentary rocks from the Karatau, Southern Kazakhstan. Stepwise thermal demagnetization reveals the presence of a high-temperature component of magnetization in most samples. Fold tests indicate a syn-folding age of magnetic remanence acquisition at three of the five areas studied. Directional data of Devonian and Permian rocks yield a positive fold test, implying a primary magnetization. Resulting prefolding paleolatitudes for Permian and Devonian rocks show the proximity of the Karatau to Baltica during those times. Syn- and post-folding magnetizations result in paleolatitudes for Karatau, which intersect the paleolatitude curve based on the Baltica apparent polar wander path (APWP), at times, which can be correlated to major deformational events at $\sim$ 280 Ma, $\sim$ 260 Ma, and $\sim$ 230 Ma, respectively. We interpret this with complicated pattern of remagnetization events accompanying deformation, which can include syn-folding remagnetization events and areas of primary magnetic signals. Additionally, the differences between reference declinations based on the APWP for Baltica and observed declinations suggest successive counterclockwise rotational reorganization of the Karatau during the late Paleozoic to Early Mesozoic, with maximal rotation values of $\sim 65°$ with respect to Baltica. The remagnetization events are correlated with latest intracontinental stages of orogenic evolution in the Ural mountains and thus the Paleozoic amalgamation of the Eurasian continent and suggest synchronous and coherent tectonic evolution in the Urals and Karatau mountains.

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1. Introduction

[2] The Central Asian Orogenic Belt (CAOB) or Altaids (Figure 1) represents one of the largest accretionary orogens and one of the places, where formation of Phanerozoic continental crust was most excessive on the Earth [*Mossakovsky et al.*, 1993; *Windley et al.*, 2007]. It consists of heterogeneous domains integrating Precambrian microcontinents, island arcs and oceanic domains, ophiolite complexes, subduction- and collision-related metamorphic complexes, and ancient accretionary wedges, which were welded together from Neoproterozoic to Permian times. A variety of tectonic models consider the CAOB as a result of interaction of numerous short-living arcs separated by diachronous sutures and general reconstructions resemble present-day setting of the west Pacific-Indonesia region [*Zonenshain et al.*,1990, *Mossakovsky et al.*, 1993, *Berzin and Dobretsov*, 1994, *Yakubchuk et al.*, 2001, *Buslov et al.*, 2002, *Windley et al.*, 2007, *Xiao et al.*, 2008, 2010]. An alternative model argues in favor of a primarily continuous single arc which was deformed by giant strike-slip faults [*Şengör et al.*, 1993], though in many cases this model does not explain the diversity of tectonic environments and the nature of various structural episodes known in different parts of the CAOB [*Windley et al.*, 2007].

[3] It was already assumed in the earliest models based on geological evidence that the western part of CAOB within Kazakhstan represents a major orocline [*Zonenshain et al.*, 1990] in the southern prolongation of the Uralides. Oroclinal bending was recently confirmed by paleomagnetic data between North Tianshan and Chingiz Range [*Levashova*

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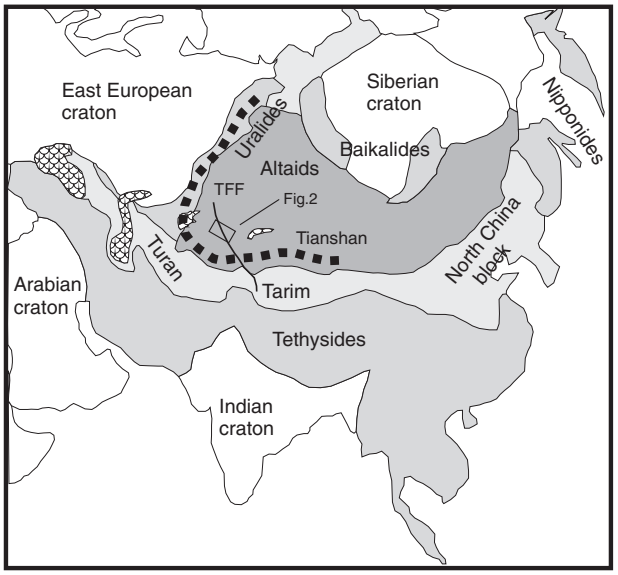


Figure 1. Simplified tectonic map of Eurasia. Thick dashed line shows Ural-Tianshan belt. TFF - Talas-Ferghana Fault. Outline of Figure 2 is marked with black rectangle.

et al., 2007, *Abrajevitch et al.*, 2007, 2008]. Extensive paleomagnetic studies during last decade had also significantly promoted the understanding of the general paleogeographic evolution of the western CAOB [*Bazhenov et al.*, 1999, 2003; *Alexyutin et al.*, 2005; *Van der Voo et al.*, 2006; *Levashova et al.*, 2007; *Abrajevitch et al.*, 2007, 2008] although a number of questions are still far from solved. The principal uncertainties are connected with (a) the early Paleozoic histories of individual terrains and their travel paths prior to formation of the Kazakhstan continent, (b) the motion of the Kazakhstan continent during middle and late Paleozoic time and mechanisms of its incorporation in Eurasia, and (c) the overall kinematics of the Late Paleozoic Kazakhstan and relative role of rotations, bending and wrenching prior to and during a process of multiple collisions of Kazakhstan with Siberia, Baltica and Tarim in the late Carboniferous and Permian, synchronous with the Uralian orogen .

[4] Our current paleomagnetic study is devoted to Devonian, Carboniferous, and Permian sedimentary rocks in Karatau Range of Southern Kazakhstan. This area is located at a junction of two Paleozoic collisional thrust and fold belts of the Urals and the pre-Cenozoic Tianshan, which delineate western and southern boundaries of the CAOB. It provides important hints for a better understanding of above two belts as well as for a reconstruction of the entire CAOB, but remains so far significantly less studied in comparison to adjacent regions. The results of our study reveal major rotations due to syn-collisional strike slip motions and allow us to establish a link between remagnetization and tectonic events.

2. Geological Setting and Tectonic History

[5] The Karatau ("Black mountains") of South Kazakhstan is a mountain range up to a height of 2 km, which trends NW-SE between the Syr-Dar'ya and Chu-Sarysu basins (inset in Figure 2) and consists mainly of Neoproterozoic and Paleozoic rocks (Figure 2). Structurally, it is subdivided into the Greater (Bolshoi) and Lesser (Malyi) Karatau, separated from each other by the Main Karatau Fault (MKF), which represents a Northern segment of the Talas-Ferghana Fault (TFF) [*Abdulin et al.*, 1986; *Allen et al.*, 2001; *Rolland et al.*, 2013]. Its geological and structural setting has been described by a number of authors [*Abdulin et al.*, 1986; *Cook et al.*, 1991, 1995; *Allen et al.*,2001; *Cook et al.*, 2002; *Alexeiev et al.*, 2009]. After Neoproterozoic

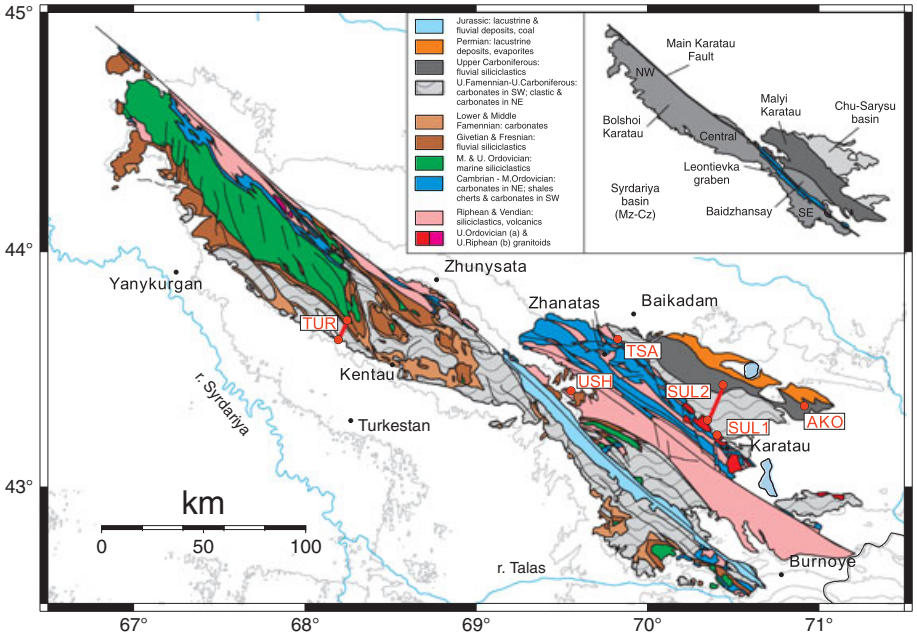


Figure 2. Geologic map of the Karatau Range (modified after *Alexeiev et al.* [2009]). Small: tectonic setting of the Karatau Range. Red dots indicate sampled areas.

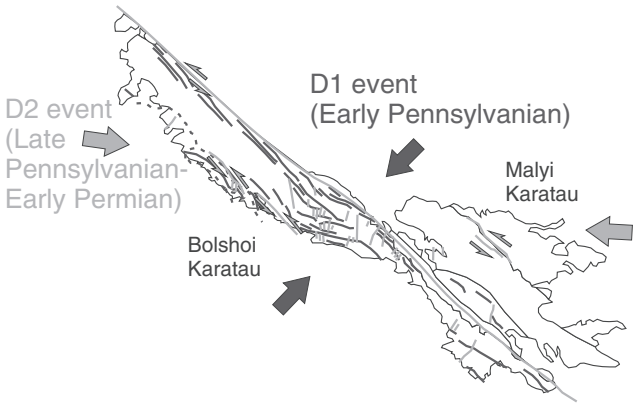


Figure 3. Distribution of characteristic structures and shortening directions of two principal deformation events. Early Pennsylvanian event at ∼ 315 Ma (dark grey). Late Pennsylvanian to early Permian event, around ∼ 300 Ma (light grey). Solid lines show fold axes, lines with arrows show strike slip faults (modified after *Alexeiev et al.* [2009]).

rifting in the Karatau, several carbonate seamounts developed due to thermal subsidence of the newly formed crust. This sedimentation was terminated by late Ordovician compressional deformation, related to amalgamation of several terrains in the Kazakhstan continent. Extensive transgression in the Middle Devonian was followed by formation of a major passive margin carbonate platform during the Famennian to early Pennsylvanian within Bolshoi Karatau. During the same time, shallow marine epi-continental basins with mixed carbonate and clastic facies evolved in the NE within the Chu-Sarysu basin [*Cook et al.*, 2002]. Carbonate sedimentation in the Bolshoi Karatau ceased in middle Bashkirian (early Pennsylvanian) as a result of the collision of Kazakhstan with Baltica (East European craton), and orogenic uplift in the Karatau and Tianshan isolated the Chu-Sarysu basin from the ocean. Since middle Bashkirian to late Permian, this terrain evolved as an intracontinental basin with fluvial and lacustrine deposits [*Abdulin et al.*, 1986; *Allen et al.*, 2001].

[6] Recent structural studies in the Karatau have revealed three distinct phases of deformation during the Pennsylvanian to early Mesozoic [*Alexeiev et al.*, 2009]. The first episode (D1) in the early Pennsylvanian is characterized by NW-trending folds and thrusts with a general direction of motion towards NE (Figure 3). The second episode (D2), which took place presumably during the late Pennsylvanian to early Permian, reflects shortening in east-west direction, evidenced by N-S trending folds and NW trending sinistral strike slip faults (Figure 3). The third episode (D3) was dominated by shortening in a N-S direction connected with the initiation of dextral motion along the Talas Ferghana Fault [*Alexeiev et al.*, 2009]. The age of the latter can be constrained within late Permian to early Mesozoic, based on recent Ar-Ar age data obtained for Talas-Ferghana fault [*Rolland et al.*, 2013].

[7] All three phases of deformation can be correlated with synchronous orogenic increments in the Uralian belt following amalgamation of Baltica, Siberia and Kazakhstan paleocontinents in Eurasia [*Echtler et al.*, 1996, 1997; *Brown et al.*, 2008].

3. Local Geologic Setting and Sampling

[8] During the field campaign, 434 oriented samples were drilled from 71 sites in six areas within Malyi and Bolshoi Karatau (Figure 2) covering Devonian, Carboniferous, and Permian sedimentary rocks. Orientation of samples and bedding planes were measured using a standard magnetic compass as well as sun compass techniques. All samples were treated with thermal demagnetization techniques using a Schoenstedt oven with peak temperatures of ∼ 700°C. Magnetization directions were measured with a 2-G squid magnetometer in three components in a magnetically shielded room. Sample mean directions were calculated using at least four consecutive demagnetization steps applying the least squares method proposed by *Kirschvink* [1980] on linear portions of the demagnetization paths.

[9] Rock magnetic parameters of representative samples were obtained using a variable field translation balance (VFTB, *Krasa et al.* [2007]) and the Lowrie method [*Lowrie*, 1990], to get information about the magnetic carriers of the remanent magnetization of the sampled material.

3.1. Bolshoi Karatau

[10] Along the Zhankurgan river in the SW Bolshoi Karatau (Figure 2), one of the most complete sections in the area with rocks ranging in age from Middle Devonian (Givetian) to Pennsylvanian, early Bashkirian [*Abdulin et al.*, 1986; *Cook et al.*, 2002] is exposed. Here, Devonian and Carboniferous rocks are folded in major NW-trending synclines and anticlines, which were formed during the main deformation episode (D1) in the early Pennsylvanian [*Allen et al.*, 2001; *Alexeiev et al.*, 2009]. In the Zhankurgan valley (section TUR), 133 drill cores were obtained at 22 sites, covering upper Devonian (Frasnian) red sandstones and uppermost Devonian (Frasnian and Famennian) to lower Carboniferous (Tournaisian and Visean) carbonates, (see *Cook et al.* [2002] for detailed stratigraphic descriptions).

3.2. Malyi Karatau

[11] Five spatially separated locations were studied in the Malyi Karatau (Figure 2) including four sections of Devonian and Carboniferous rocks (TSA, USH, SUL1 and SUL2) and one section with Permian rocks (AKO).

[12] Devonian rocks sampled in the Ushbas area (section USH) in the NW Malyi Karatau (Figure 2) are part

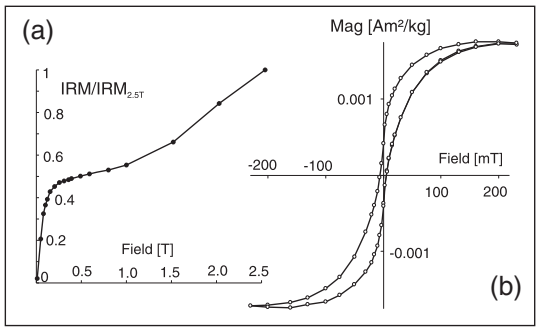


Figure 4. (a) Exemplary plot of acquisition curve of isothermal remanent magnetization (IRM) and (b) measured hysteresis loop of representative samples.

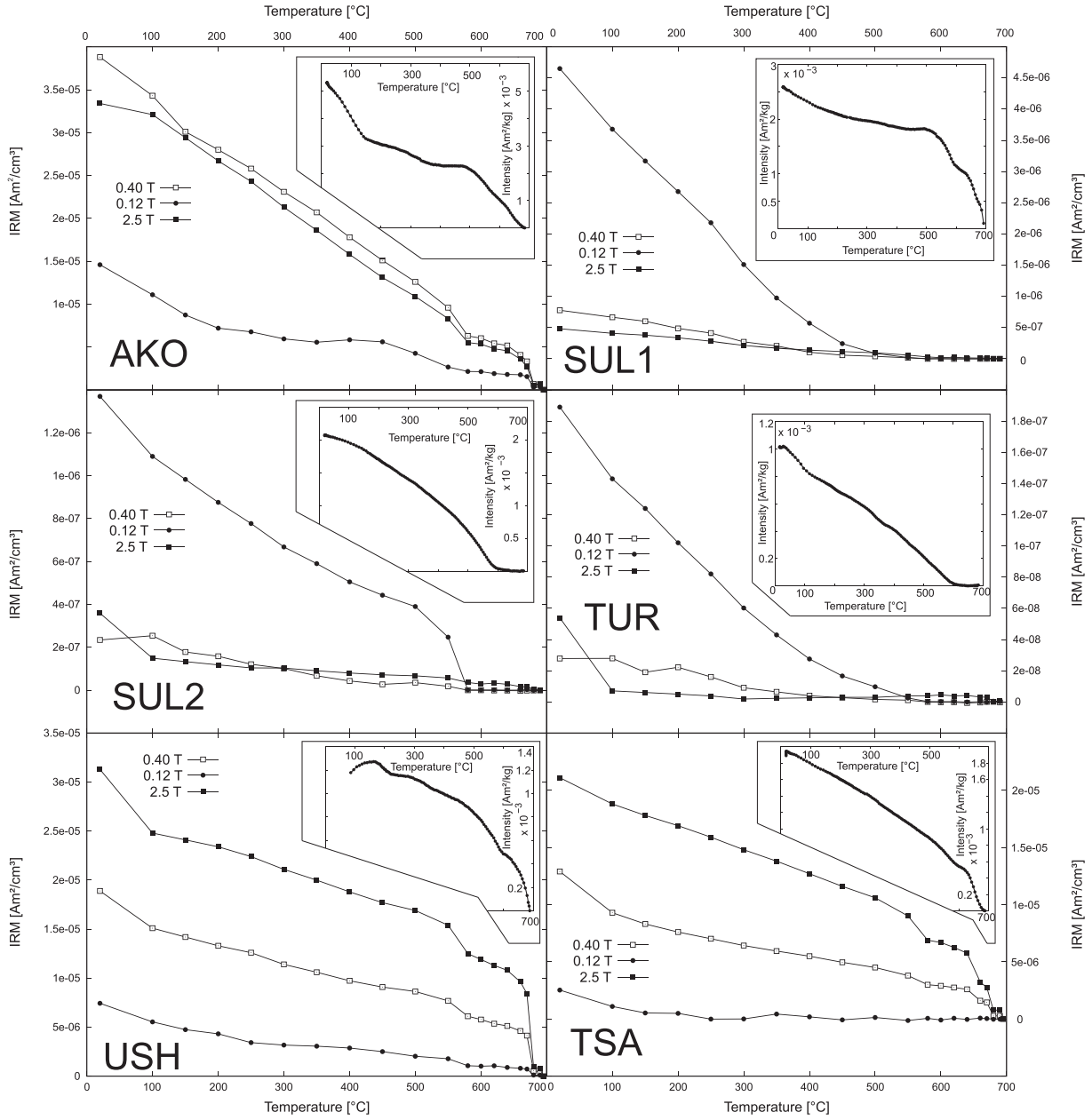


Figure 5. Thermal demagnetization of a three-axis composite IRM for one representative sample from all areas. Orthogonal DC fields of 2.5, 0.4, and 0.12 T were applied sequentially to all samples prior to heating. Also shown are representative thermomagnetic curves (with DC bias field of $\sim$ 100 mT) of rocks of the particular area (small).

of the same sedimentary formations as in the adjacent Bolshoi Karatau including Givetian and Frasnian red beds and Famennian limestones. Similar structural patterns in both the Zhankurgan and Ushbas areas suggest that the rocks were deformed during the same episode and thus a late Carboniferous age of deformation can be inferred in the Ushbas area as well. In the southern part of the Ushbas syncline, 11 sites with 67 samples were studied in middle and upper Devonian red sandstones of the Tyul'kubash formation.

[13] The section TSA is located on the road between Zhanatas and Baikadam (Figure 2) at the SW edge of Chu-Sarysu basin. The rocks comprise uppermost Devonian to lower Carboniferous red sandstones and conglomerates of the Suleimansay formation, which are deformed in a narrow NW- trending syncline [*Abdulin et al.*, 1986]. Two sites, each with six samples, were collected in fine-grained red sandstones from both limbs of the fold.

[14] Two sections were also studied in the SE Malyi Karatau on the SW flank of the Chu-Sarysu basin. In total, 14 sites with 84 samples were drilled from the section SUL-1 near the Suleimansay mine (Figure 2) from the same formation as at TSA. The sequence is dominated by brown sandstones with interlayers of light dolomites and grey limestones. The latter contain rare brachiopoda and foraminifera fossils, which constrain the age of the Suleimansay formation and the overlying sediments as latest Devonian to

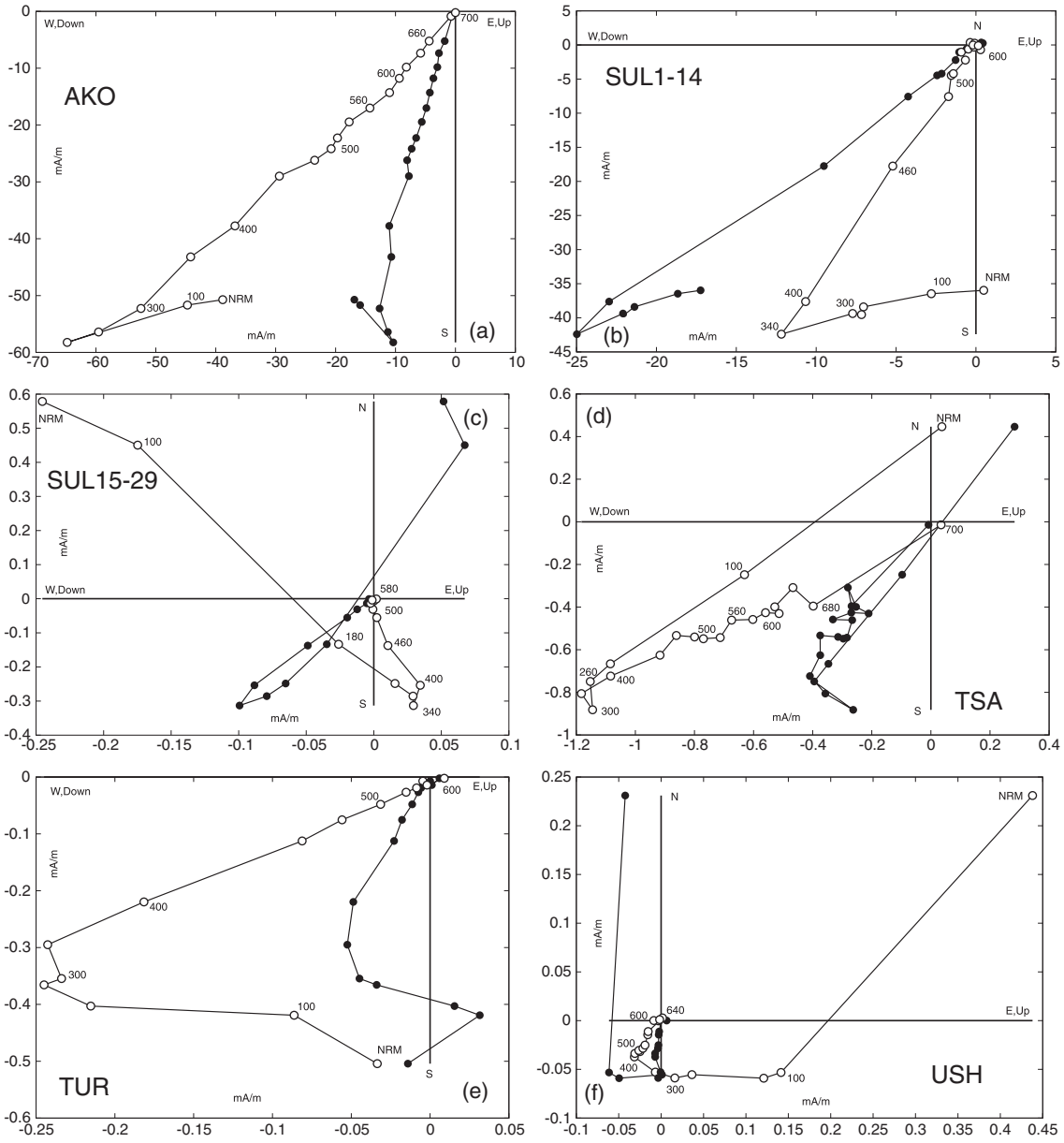


Figure 6. Results of thermal demagnetization experiments plotted as orthogonal vector diagrams [*Zijderveld*, 1967] in stratigraphic coordinates. Solid and open dots represent vector endpoints projected onto the horizontal and vertical planes, respectively. Positive values represent projection in north, east and up direction, respectively. Temperature steps in °C are indicated.

earliest Carboniferous [*Abdulin et al.*, 1986; *Cook et al.*, 2002]. At this location, 14 sites with a total of 84 samples were drilled.

[15] Section SUL-2 is exposed along the river Koktal and includes Visean, Serpukhovian, and lower Pennsylvanian rocks. The lower Carboniferous sequence is generally dominated by light-colored shallow marine limestones and fossiliferous limestones, which contain subordinate evaporites in the lower Visean, argillaceous limestones and marls in the middle Visean and sandy limestones and limy sandstones in the upper Visean and Serpukhovian. The lower Bashkirian consists mainly of shallow marine carbonate sandstones and sandy limestones with conglomerate layers. They are overlain with erosional contact by fluvial conglomerates and sandstones of the Kyzylkanat Formation, which has a Pennsylvanian age. Lower and upper Carboniferous rocks from the SUL-2 section are deformed in relatively low-angle linear folds, trending NW-SE [*Abdulin et al.*, 1986; *Allen et al.*, 2001; *Cook et al.*, 2002].Fourteen sites were investigated along the Koktal river with a total of 91 samples.

[16] The section AKO includes several sites within Permian sedimentary succession, which were sampled in the area around the town of Akkol (Figure 2) in the very east of Malyi Karatau, within the southwestern part of the Chu-Sarysu basin. Permian rocks are represented by lacustrine facies and consist of red and grey fine-grained sandstones

Table 1. Site Mean Directions for Zhankurgan Section (Givetian-Visean)

			In Situ				Bedding-Corrected			
Site	Bed.	N	D (°)	I (°)	k	α_{95} (°)	D(°)	I (°)	k	α_{95} (°)
Zhankurgan (TUR)										
Tur-1[a]	210/42	6/6	224.0	–33.6	3.1	45.7	246.5	–72.3	3.1	45.7
Tur-2[a]	233/30	6/6	181.3	–11.0	6.4	28.9	172.4	–25.8	6.4	28.9
Tur-3[a]	213/40	7/7	197.8	–5.3	4.5	35.8	190.4	–42.2	4.5	35.8
Tur-4	223/44	6/6	200.1	–15.5	21.8	14.9	181.4	–51.6	21.8	14.9
Tur-5	233/45	6/5	203.4	–28.5	140.8	6.5	167.3	–58.1	140.8	6.5
Tur-6	230/47	6/6	206.2	–29.8	302.2	3.9	165.3	–63.5	302.2	3.9
Tur-7	244/45	6/6	211.8	–26.4	135.6	5.3	178.3	–55.0	135.6	5.3
Tur-8[a]	201/39	7/7	55.9	–53.6	3.0	42.3	44.1	–17.6	3.0	42.3
Tur-9[b]	227/43	6/5	14.4	49.7	49.3	11.0	301.8	65.1	49.3	11.0
Tur-10	245/20	6/5	199.1	–34.0	77.4	8.8	186.8	–43.6	77.7	8.7
Tur-11	004/07	6/3	195.3	–39.2	68.2	15.0	194.8	–32.2	68.2	15.0
Tur-12	225/55	6/4	206.9	–7.3	35.9	15.5	187.0	–54.9	35.4	15.7
Tur-13	233/49	6/6	212.8	–11.6	15.6	18.1	193.8	–53.2	15.6	18.1
Tur-14	243/50	6/6	205.6	–9.6	146.6	5.6	187.2	–42.0	160.5	5.3
Tur-15	223/43	6/6	196.7	–34.1	57.4	9.1	155.5	–63.2	57.4	9.1
Tur-16	047/54	6/6	153.4	–56.9	67.7	8.3	196.2	–23.9	67.7	8.3
Tur-17	214/66	6/4	204.4	11.6	167.3	7.1	195.5	–51.6	167.3	7.1
Tur-18	215/25	5/5	200.9	–18.8	67.6	9.4	195.3	–42.1	67.6	9.4
Tur-19	215/75	5/5	202.1	12.0	120.3	7.0	185.7	–57.7	120.3	7.0
Tur-20	052/37	7/6	187.4	–34.2	181.9	5.0	197.9	–9.2	181.3	5.0
Tur-21	205/48	6/6	204.8	–10.5	129.9	5.9	200.3	–58.3	131.7	5.9
Tur-22	196/42	6/5	201.0	–5.5	176.9	5.8	209.1	–47.5	176.9	5.8
MEAN	–	22/18[c]	201.1	–20.2	15.6	9.3	188.4	–48.2	24.5	7.3

Site: sampling site, Bed: bedding orientation, dip direction/plunge (ss), N: number of measured samples/ number of samples used for calculating site mean directions. Declination (D) and Inclination (I) in degrees, Fisher radius (α_{95}) of 95% confidence, Fisher precision parameter k [*Fisher*, 1953], in geographic (in situ) and bedding corrected coordinates, respectively.

[a]Site mean direction not included in calculating area mean direction, because of α_{95} values larger than 20°.

[b]Rejected site because of proximity of *in situ* direction to present day field.

[c]For area mean directions: N: number of obtained site mean directions/number of used site mean directions for calculating area mean.

and siltstones with subordinate interlayers of carbonates and evaporites [*Abdulin et al.*, 1986; *Bazhenov et al.*, 1995]. Eight localities have been sampled with a total of 47 samples.

4. Rockmagnetic Results

[17] In order to obtain information on the magnetic inventory, representative samples from all different areas were studied rockmagnetically using a Variable Field Translation Balance (VFTB, *Krasa et al.* [2007]) to obtain magnetic hysteresis data, isothermal remanent magnetization (IRM), thermomagnetic curves, and thermal demagnetization of a three-axis composite IRM. In general, none of the samples subjected to IRM experiments saturates at 2.5 T, and the IRM acquisition curves indicate the presence of at least two magnetic phases with different coercivity spectra (Figure 4a). Furthermore, hysteresis loops are either characterized by pot-bellied or wasp-waisted shape [*Tauxe et al.*, 1996], which is also an argument for the existence of more than one prominent magnetic mineral (or a mixture of different grain sizes) in these rocks (Figure 4b).

[18] Additional information was acquired using thermal demagnetization of composite three-axis IRMs [*Lowrie*, 1990] in combination with in-field ($\sim$ 100 mT) thermomagnetic curves in (insets in Figure 5). The orthogonal IRMs were imparted sequentially using DC fields of 2.5, 0.4, and 0.12 T.

[19] The Permian red sandstones (AKO) are characterized by unblocking temperatures of the natural remanent magnetization (NRM) of $\sim$ 680 – 700°C. Furthermore, there are indications for additional magnetic phases, which is, on the one hand, a low temperature phase up to $\sim$ 200°C, visible both in thermomagnetic curves and when demagnetizing the low coercivity IRM. Additionally, the IRM demagnetization shows distinct unblocking temperatures of $\sim$ 580°C and $\sim$ 680°C for all rocks, independent of coercivity (Figure 5).

[20] Rocks from Suleimansay (SUL-1 and SUL-2) and Zhankurgan (TUR) show prominent unblocking temperatures at $\sim$ 550 – 580°C in thermomagnetic curves and in the low coercivity IRM (Figure 5). SUL-1 shows an additional high temperature phase at the (in-field) thermomagnetic curve, which drops to zero at $\sim$ 680°C.

[21] Samples from Ushbas and Zhanatas indicate two unblocking temperatures at $\sim$ 580°C and $\sim$ 680°C, respectively, at both thermomagnetic curves and at demagnetization of orthogonal IRMs (Figure 5).

[22] In conclusion, these results suggest the predominance of magnetite together with differing amounts of hematite within all the samples. In conclusion, the magnetomineralogy of all sampled lithologies is dominated by magnetite. The magnetic properties of samples from AKO,

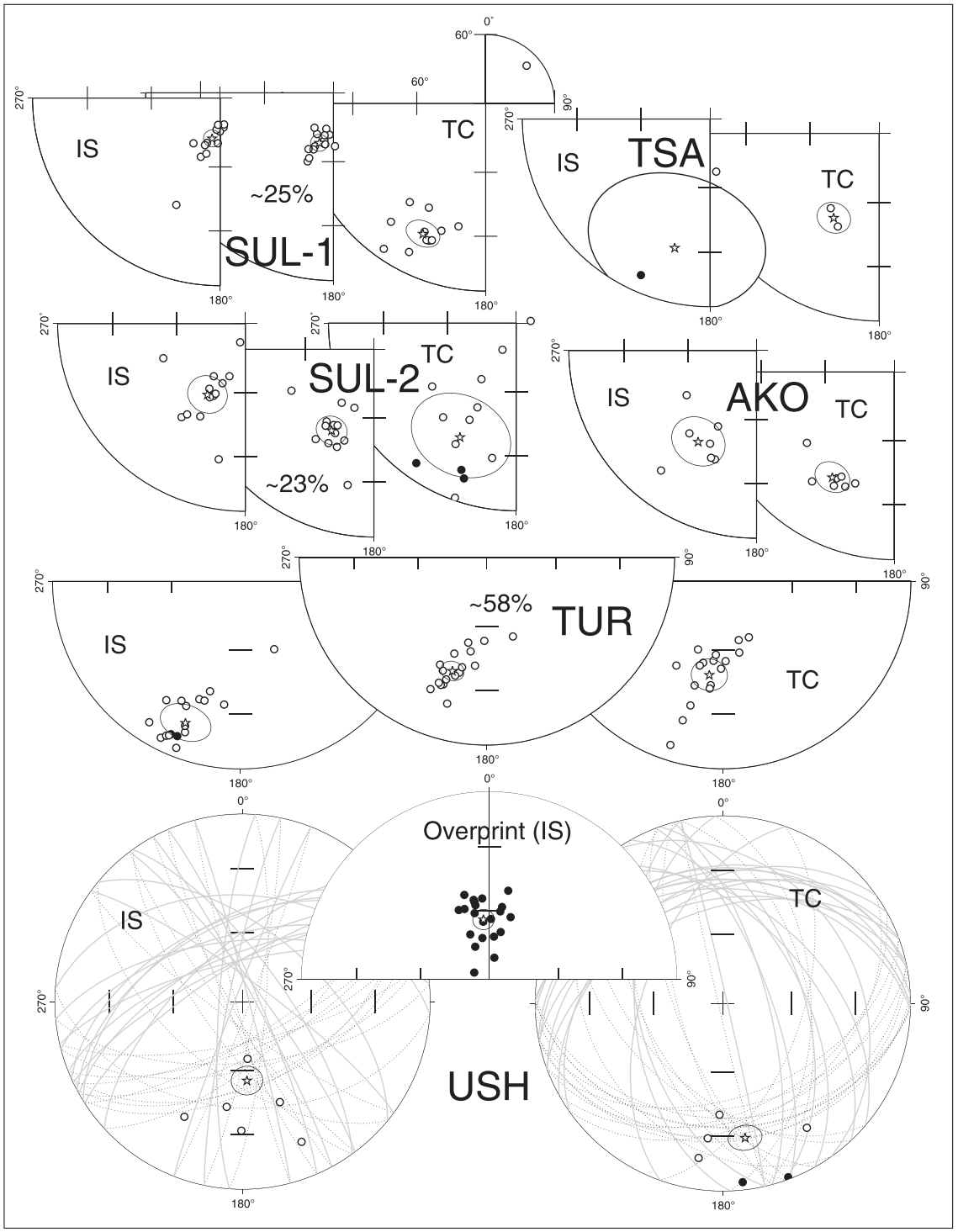


Figure 7. Stereographic projections of site mean directions for all sampled areas located in the Bolshoi and Malyi Karatau. IS refers to in situ data, and TC refers to tilt-corrected data. The stars indicate the area mean direction with 95% confidence circles. Solid and open dots represent projection on the upper and lower hemispheres, respectively. Central plot shows, where present, mean directions after % of unfolding for maximal k value [*Watson and Enkin*, 1993]. USH: overprinted component of representative samples is also shown in in situ coordinates.

Table 2. Site Mean Directions for Zhanatas (Famennian) and Akkol (Permian) Sections

			In Situ				Bedding-Corrected			
Site	Bed.	N	D (°)	I (°)	k	α_{95} (°)	D (°)	I (°)	k	α_{95} (°)
Zhanatas (TSA)										
Tsa-1	064/28	6/4	173.3	–67.1	612.2	3.7	212.9	–50.9	612.2	3.7
Tsa-2	196/55	6/4	203.9	10.0	36.7	15.4	203.9	–45.0	36.7	15.4
MEAN	–	2/2[a]	195.3	–29.7	3.6	34.1	208.2	–47.7	66.3	6.9
Akkol (AKO)										
Ako-1	258/25	6/6	218.5	–19.5	28.1	12.9	208.4	–35.9	28.1	12.9
Ako-2	055/05	5/5	200.9	–38.1	270.1	4.7	203.1	–34.1	270.1	4.7
Ako-3[a]	030/04	6/6	149.0	–71.0	3.9	39.0	158.1	–68.8	3.9	39.0
Ako-4	300/05	6/6	203.0	–38.1	152.1	5.4	199.2	–37.3	152.1	5.4
Ako-5	025/15	6/4	206.9	–52.7	65.8	11.4	206.7	–37.7	61.7	11.8
Ako-6	025/15	7/7	237.5	–53.7	28.4	11.5	230.9	–40.0	28.4	11.5
Ako-7	025/15	5/3	219.0	–42.8	71.4	14.7	216.9	–28.1	65.8	15.3
Ako-8	035/12	6/6	206.5	–44.0	1193.2	1.9	207.9	–32.1	1156.3	2.0
MEAN	–	8/7[a]	212.6	–41.9	31.1	11.0	210.4	–35.3	74.9	7.0

Site mean direction not included in calculating area mean direction, because of large α_{95} values.
[a]For area mean directions: N: number of obtained site mean directions/number of used site mean directions for calculating area mean.

USH, TSA, and, to a lesser degree, from SUL1 show a significant contribution of hematite.

5. Demagnetization Results

[23] In accord with our rockmagnetic results, detailed thermal demagnetization experiments reveal the presence of at least two magnetic components, with different paleomagnetic directions. An initial low temperature component (LTC) was identified during thermal stepwise demagnetization between 100°C and up to 300°C. LTC is broadly pointing to the North and down and yields an overall mean direction, which is consistent with the present day magnetic field in the sampling area. Only in samples from Ushbas the LTC is present up to ∼ 600°C.

[24] A high temperature component (HTC) of reversed polarity in tilt corrected coordinates was identified as linear segments from ∼ 300–350°C up to ∼ 600–680°C. The samples with unblocking temperatures larger than 600°C carry two high temperature components, which are, however, not distinguishable within error limits (Figures 6a and 6d).

5.1. Bolshoi Karatau

[25] *Zhankurgan:* The LTC of Zhankurgan area (TUR) yields a mean direction of D = 1.9°, I = 61.7°, k = 31.5 and α_{95} = 2.6° (*in situ*) based on 92 samples. Furthermore, a high temperature component (HTC) is identified in 91% (121 specimens) of all treated specimens with unblocking temperatures of up to ∼ 600°C. Based on sample mean directions of the HTC site, mean directions were calculated (Table 1). Five of the 22 site mean directions are rejected because of α_{95} values higher than 20° for further analysis. The remaining 17 sites mean that directions were subjected to the fold test of *Watson and Enkin* [1993]. Stepwise unfolding results in a maximum k-value at ∼ 58% untilting with a confidence interval of 5.2° (Figure 8), indicating an acquisition age of the characteristic remanence during folding. The resulting syn-folding area mean direction is D = 196.9°, I = –37.0°, k = 60.4, and α_{95} = 4.6° (Figure 7).

5.2. Malyi Karatau

[26] *Zhanatas:* A LTC is identified in 11 samples from Zhanatas area and yields a mean direction of D = 14.4°, I = 58.1°, k = 34.7, and α_{95} = 7.9° in in situ coordinates. Averaging the HTC from 12 samples of the two sites results in D = 190.8°, I = –39.5°, k = 3.6, and α_{95} = 30.0° before and D = 207.1°, I = –48.0°, k = 11.9, and α_{95} = 14.6° after tilt correction (Figure 7 and Table 2). Stepwise unfolding indicates a maximum k-value at 100% unfolding (Figure 8), which suggests a prefolding age of this magnetization. Calculating mean directions and performing fold tests on only two site mean directions would not yield statistically sufficient results. Therefore, all statistic procedures were carried out on sample level.

[27] All treatments are conducted with sample mean directions. Unfortunately, outcrop conditions in this area have not allowed a more detailed sampling.

[28] *Suleimansay:* Thermal demagnetization experiments of samples from area SUL-1 indicate an LTC component below 300°C (Figure 6), which yields a sample mean direction of D = 9.1°, I = 59.8°, k = 34.9, and α_{95} = 3.0° (in situ) based on 66 samples. Above 300°C, a high temperature component (HTC) is observed up to ∼ 680°C (Figure 6 and Table 3). Stepwise unfolding of these site mean directions results in a maximum k value after ∼ 25% unfolding (Figure 8). Performing the fold test of *Watson and Enkin* [1993] leads to a maximum k-value at 25.4% with an error of ±8.1%. The syn-folding direction with the maximum k-value is D = 198.5°, I = –60.4°, k = 134.5, and α_{95} = 3.6° (Figure 7).

[29] At SUL-2, 85 of the 91 samples show stable demagnetization behavior, revealing the presence of both a LTC and a HTC. Stepwise heating up to ∼ 300°C completely removes the LTC. The sample mean direction for LTC is

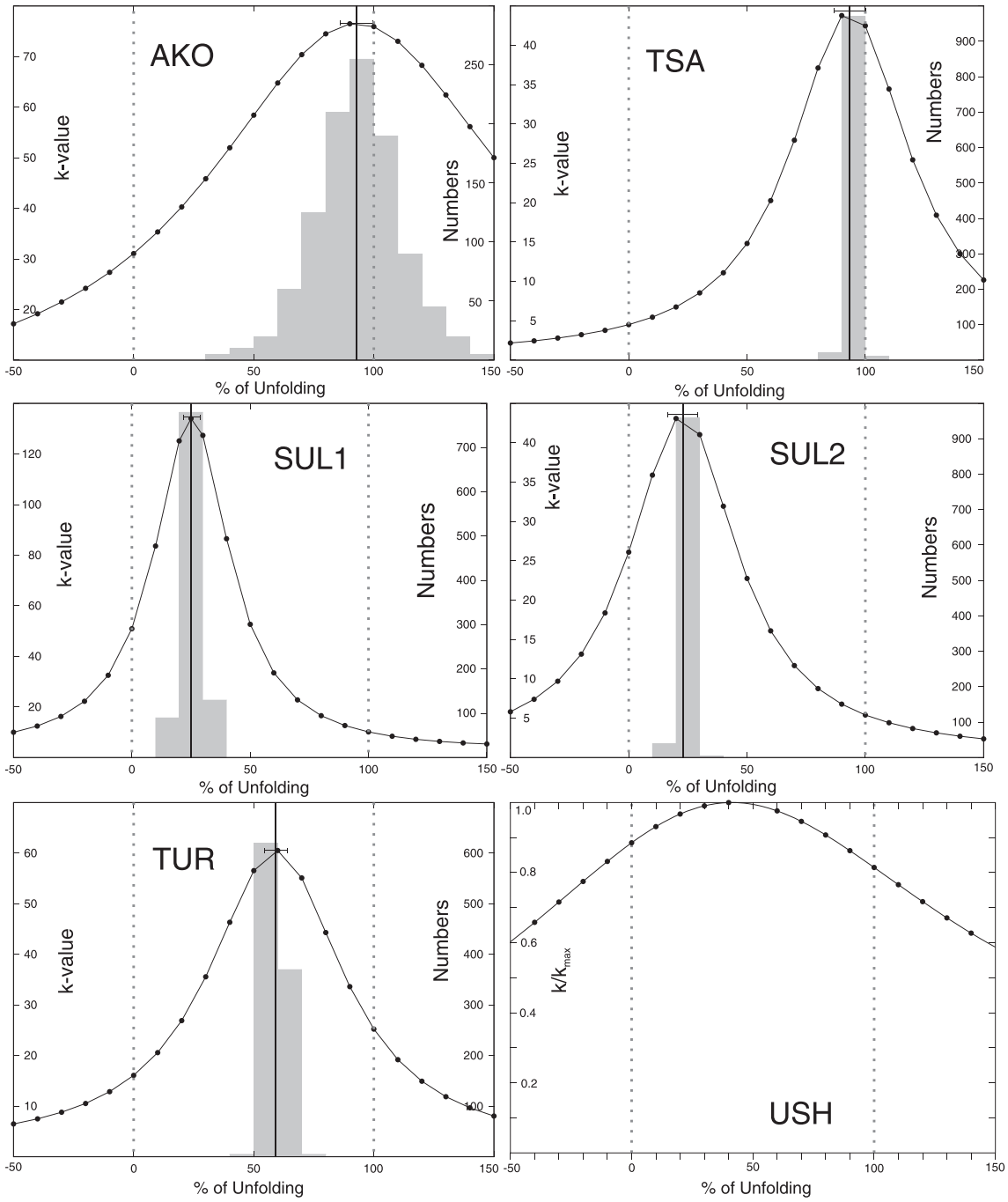


Figure 8. Plots of precision parameter k for stepwise unfolding of site mean directions for every area. Dotted lines indicate 100% and 0% unfolding, respectively. Black line shows maximum k-value with error limit, which is obtained from 1000 parametric simulations *Watson and Enkin* [1993]. The results of these simulations are shown in grey histograms. Area USH shows k-values for stepwise unfolding including remagnetization great circles [*McFadden*, 1990].

D = 2.6°, I = 61.6°, k = 82.8, and α_{95} = 1.9° based on 85 samples. Between ∼ 300°C and ∼ 600°C, a HTC is present. Site mean directions of 14 sites yield a mean directions of D = 208.1°, I = –54.8°, k = 26.1, and α_{95} = 8.3 *in-situ* and D = 206.2°, I = –32.8°, k = 5.4, and α_{95} = 19.7° after tilt correction (Figure 7 and Table 4). We note, however, that a maximum k-value is reached at ∼ 23% unfolding (Figure 8), which indicates a syn-folding age of remanence acquisition and results in a mean direction of D = 207.5°, I = –49.8°, k = 43.6, and α_{95} = 6.4° (Figure 7). The statistical fold test of *Watson and Enkin* [1993] yields a narrow interval of unfolding percentages (±3.5%). This confirms a robust syn-folding age of the magnetization acquisition.

[30] *Ushbas:* The LTC of Ushbas area (USH) results in a mean direction of D = 5.5°, I = 66.0°, k = 14, and

Table 3. Site Mean Directions for Suleimansay-1 Section (Famennian-Visean)

			In Situ				Bedding-Corrected			
Site	Bed.	N	D (°)	I (°)	k	α_{95} (°)	D (°)	I (°)	k	α_{95} (°)
Suleimansay-1 (SUL-1)										
Sul-1	022/50	4/3	173.4	–77.3	191.2	8.9	199.2	–29.1	180.4	9.2
Sul-2[a]	025/45	4/4	10.0	–75.8	8.2	34.2	219.1	–58.0	8.2	34.2
Sul-3	026/45	6/6	196.5	–69.8	285.2	4.0	205.5	–25.3	285.2	4.0
Sul-4	025/55	6/6	180.6	–75.5	179.7	5.0	202.4	–22.1	179.7	5.0
Sul-5	030/52	6/6	198.2	–63.1	477.7	3.1	207.3	–12.0	477.7	3.1
Sul-6	022/47	6/6	197.9	–68.6	399.8	3.4	203.4	–21.8	399.8	3.4
Sul-7	040/37	5/4	182.9	–70.4	686.2	3.5	208.6	–37.0	686.3	3.5
Sul-8	200/75 215/75	6/5	202.3	–38.5	54.3	10.5	47.3	–65.9	26.3	15.2
Sul-9	033/74	8/7	194.0	–65.2	38.3	9.9	215.0	–6.1	38.3	9.9
Sul-10	055/55	6/5	171.8	–78.5	357.3	4.1	218.7	–20.4	357.3	4.1
Sul-11	040/45	8/8	186.7	–76.3	137.5	4.7	217.5	–34.2	136.4	4.8
Sul-12	010/45	6/6	167.4	–71.7	19.1	18.0	185.5	–28.3	19.1	18.0
Sul-13	010/45	6/6	210.6	–67.2	96.0	6.9	201.5	–22.8	96.0	6.9
Sul-14	010/45	7/7	184.4	–78.4	22.1	13.1	192.5	–33.5	22.1	13.1
MEAN	–	14/13[a]	191.3	–72.2	151.3	3.7	205.70	–24.0	51.3	6.4

Site mean direction not included in calculating area mean direction, because of large α_{95} values.
[a]For area mean directions: N: number of obtained site mean directions/number of used site mean directions for calculating area mean.

α_{95} = 4.0 based on 67 samples. The HTC shows ambiguous behavior. Above 300°C and up to 700°C stepwise demagnetization experiments yield curved demagnetization trajectories, failing to reach the origin of the projection. Using the *McFadden and McElhinny* [1988] algorithm to analyze great circle data, including at least six stable endpoints, with mean directions of six exceptional high quality samples (Figure 7) yield a characteristic mean direction of D = 176.8°, I = –55.4°, k = 19.0, and α_{95} = 6.5° in in situ coordinates and D = 170.6°, I = –28.3°, k = 19.9, and α_{95} = 6.3° after tilt correction. Stepwise unfolding of combined directional and great circle data [*McFadden*, 1990] results in a maximum of the *Fisher* [1953] k value at ∼ 50% unfolding (Figure 8) suggesting that the resulting mean direction D = 173.0°, I = –41.5°, k = 23.5, and α_{95} of 5.8° is of syn-folding age.

[31] Since a large number of samples from all sites at Ushbas show only LTCs during demagnetization, no site mean directions for HTC could be obtained here. Our procedure of computing an area mean direction out of samples from different sites certainly enlarges the error, which is not sufficiently represented by the α_{95} value.

[32] *Akkol:* The LTC holds a mean direction of D = 358.1°, I = 64.2°, k = 31.5, and α_{95} = 4.2° in in situ coordinates, using data of 38 samples. Between 300°C and ∼ 700°C there is a HTC present. Based on mean directions of individual sites, an area mean direction is calculated. The other seven sites yield a mean direction of D = 212.6°,

Table 4. Site Mean Directions for Suleimansay-2 Section (Serpukhovian-Moscovian)

			In Situ				Bedding-Corrected			
Site	Bed.	N	D (°)	I (°)	k	α_{95} (°)	D (°)	I (°)	k	α_{95} (°)
Suleimansay-2 (SUL-2)										
Sul-15	030/26	6/6	203.7	–55.6	181.4	5.0	206.6	–28.8	181.4	5.0
Sul-16	215/55	7/7	214.3	–39.6	37.1	10.7	81.7	–83.7	37.1	10.7
Sul-18	038/68	6/6	208.9	–57.6	701.9	2.5	215.5	9.3	701.9	2.5
Sul-19	060/24	5/5	203.0	–56.8	149.2	6.3	217.8	–36.8	149.2	6.3
Sul-20	355/15	6/4	247.2	–51.0	125.7	8.7	233.0	–43.4	125.7	8.7
Sul-21[a]	227/48	6/5	214.3	-54.8	11.7	20.7	91.7	–68.5	11.7	20.7
Sul-22	253/02	6/6	191.1	–27.3	63.4	8.5	190.1	–28.0	63.4	8.5
Sul-23	210/35	8/8	212.7	–42.2	211.1	3.8	207.3	–77.1	211.1	3.8
Sul-24	040/02	6/6	210.9	–63.4	172.8	5.1	211.8	–61.4	172.8	5.1
Sul-25	015/80	5/5	195.4	–81.7	29.9	14.2	199.3	–1.7	29.9	14.2
Sul-26	015/80	6/5	201.0	–62.3	150.2	6.3	200.5	17.7	150.2	6.3
Sul-27	015/80	7/6	196.7	–66.2	163.7	5.3	198.6	13.8	163.7	5.3
Sul-28	010/05	9/8	205.8	–54.2	109.1	5.3	204.7	–49.3	109.1	5.3
Sul-29	013/02	8/8	205.9	–44.4	183.9	4.1	205.7	–42.4	183.3	4.1
MEAN	–	14/13[a]	208.1	–54.8	26.1	8.3	206.2	–32.8	19.7	5.4

Site mean direction not included in calculating area mean direction, because of large α_{95} values.
[a]For area mean directions: N: number of obtained site mean directions/number of used site mean directions for calculating area mean.

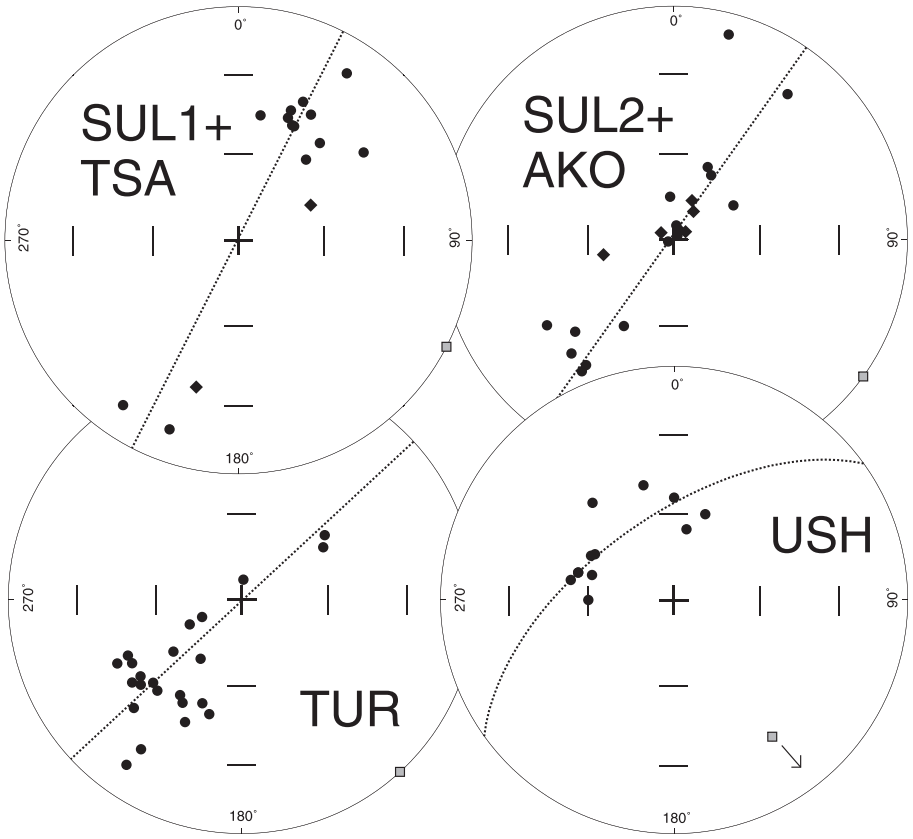


Figure 9. Stereographic projection of normal vectors of all bedding planes for respective areas. Dotted lines indicate fitted great circle. Solid dots represent projection on the upper hemisphere. Grey squares indicate resulting fold axis, arrow shows supposed direction of correction for tilted fold axis. Bedding planes of TSA and AKO are indicated as diamonds.

I = –41.9°, k = 31.1, and α_{95} = 11.0° before and D = 210.4°, I = –35.3°, k = 74.9, and α_{95} = 7.0° after tilt correction (Figure 7 and Table 2). Stepwise unfolding shows a maximum k-value at 100% unfolding (Figure 8), which suggests a prefolding age of the magnetization acquisition.

5.3. Folding Symmetry

[33] Fold tests using stepwise unfolding techniques are usually based on the assumption that the rotation of bedding planes occur about a horizontal axis (fold axis). In cases where deformation occurred by rotation about inclined or multiple axes, the true direction of syntectonic (or even pretectonic) magnetizations cannot be restored by these algorithms. In order to control these effects, the geological structures of all sampled areas were investigated in more detail.

[34] The deformation pattern in sections Akkol (AKO), Suleimansay 1 + 2 (SUL1+2), Zhanatas (TSA) and Zhankurgan is characterized by upright symmetrical folds with NW-SE trending horizontal fold axes and vertical axial planes as shown by stereo plots of normal vectors to bedding planes (Figure 9). This is taken as a strong argument that fold formation took place in a symmetrical way. Consequently, we infer that for these sections, the directions of remagnetization acquired during deformation can be correctly restored by symmetrical stepwise unfolding as applied by the fold tests used in this study [*McFadden*, 1990; *Watson and Enkin*, 1993]. At Ushbas, the axial planes of folds are equally vertical, however, the fold axes dip with ∼ 30° towards SE (Figure 9). This indicates that at Ushbas area, which is located in the vicinity of the TFF, deformation was more complex and challenges the reliability of the results of the fold test analysis.

5.4. Data Summary

[35] After removal of a LTC, which is generally aligned with the present day field, a HTC is identified in most of the thermally demagnetized material from Bolshoi- and Malyi-Karatau (Tables 1–4). Site mean directions with α_{95} greater than (20°) were rejected and not used for further analysis. The HTC directions isolated in the sections at SUL-1, SUL-2, TUR, and USH (Figures 7 and 8) indicate a syn-folding age of acquisition of the magnetic signal (Table 5), which suggests that the magnetization was acquired during a process of folding [*Bachtadse et al.*, 1987; *McFadden*, 1990]. Two area mean directions show a maximum k [*Fisher*, 1953] (AKO: 43.8 and TSA: 62.7) after tilt correction (Figure 8) indicating a prefolding age of magnetization.

[36] The area of Ushbas (USH) shows a plunging fold axis (Figure 9). The result of the syn-tilt direction is therefore questionable and might be biased by unknown amounts of rotation about inclined axes. Because of this, and also due to

Table 5. Area Mean Directions After Refolding Procedure at Maximal k Values With Resulting Paleopoles

Area	N	D (°)	I(°)	k	α_{95} (°)	(%) unf.	λ (°)	Plat	Plon	dp	dm	Inf. Age of Magn.
SUL-1	13	198.5	–60.4	134.5	3.6	25.4 ± 8.1	41.4	–76.2	341.8	5.5	4.2	~ 220 – –230
SUL-2	13	207.5	–49.8	43.6	6.4	22.8 ± 3.5	30.6	–64.8	1.5	8.5	5.7	255–260
TSA	12	207.1	–48.0	44.4	6.7	100	29.0	–64.0	4.5	19.1	12.5	355–360[b]
AKO	7	210.4	–35.3	74.9	7.0	100	19.5	–55.1	14.4	4.7	8.1	~255–290[b]
TUR	18	196.9	–37.0	60.4	4.6	58.4 ± 5.2	20.6	–63.0	31.5	3.2	5.4	275–280
USH	22,6[a]	173.0	–41.5	23.5	5.8	51	23.9	–	–	–	–	–

N: number of mean directions, used for performing fold tests [*McFadden*, 1990; *Watson and Enkin*, 1993] and calculating area mean directions, mean Declination (D), mean Inclination (I), and Fisher [*Fisher*, 1953] radius of 95% confidence (α_{95}) after (%) unfolding of the directional data. Error limits of unfolding degree is obtained using Monte Carlo simulations [*Watson and Enkin*, 1993]. λ is the paleolatitude. Resulting paleopole (pole latitude (Plat) and longitude (Plon)) with error ellipse dp and dm.

[a]At Ushbas (USH), 22 demagnetization great circles were combined with linear segments of six samples [*McFadden and McElhinny*, 1988] to get a regional mean direction.

[b]Inf. Age of Magn.: Inferred age of acquisition of the magnetization (primary ([b]) or due to remagnetization).

the poor quality of the demagnetization results, we omit this area from further analysis.

[37] The directional mean data was transformed into five paleopole positions (Table 5). The paleopoles are not consistently grouped within α_{95} error limits, nor do they fall onto any part of the Paleozoic apparent polar wander path (APWP) of Baltica [*Smethurst et al.*, 1998]. All poles are, however, rotated counterclockwise (ccw) with respect to their expected locations in vicinity to the Baltic APWP.

6. Age of Magnetization

[38] Fold tests (Figures 7 and 8) indicate that four of the six Paleozoic areas studied have been subjected to remagnetization. The sampled rocks at AKO span most of the Permian period. The positive fold test at 100% of unfolding suggests that the primary magnetization was acquired prior to Mesozoic deformation. The slight underestimation of the paleolatitude compared to expected values from the APWP of Baltica can be explained by inclination shallowing, which has been proposed for Permian red beds from South Kazakhstan [*Bazhenov et al.*, 1995] and red beds in general [e.g., *Tauxe and Kent*, 2004]. However, the syn-folding acquisition of magnetization observed in rocks from the other areas studied makes inclination shallowing rather unlikely.

[39] It is now widely accepted that remagnetization of unmetamorphosed sedimentary rocks is related to orogenically triggered fluid flow from the orogen into the foreland [*Oliver*, 1986] and the associated new formation of magnetite and to a smaller extent, hematite [*McCabe and Elmore*, 1989]. Expanding the orogenic fluid flow concept, *Zegers et al.* [2003] stress the importance of elevated

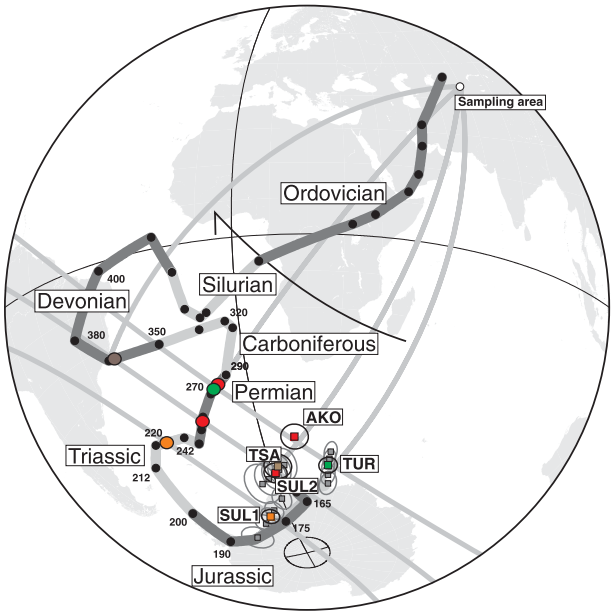


Figure 10. Colored squares indicate paleopoles (TSA - brown, TUR - green, SUL2 and AKO - red and SUL1 - orange), calculated from the characteristic area mean directions. Shown are the proposed deviations compared to the apparent polar wander path (APWP) of the East European Craton [*Smethurst et al.*, 1998] (with ages in Myrs). Wedges about the sampling area illustrate the rotational reorganization about a vertical axis. Colored ellipses indicate intersections of particular small circles with the associated part of the APWP. Grey squares at TUR, SUL1, and SUL2 show paleopoles calculated from directions for a bandwidth of ±20% around the optimal (maximum k) unfolding percentage.

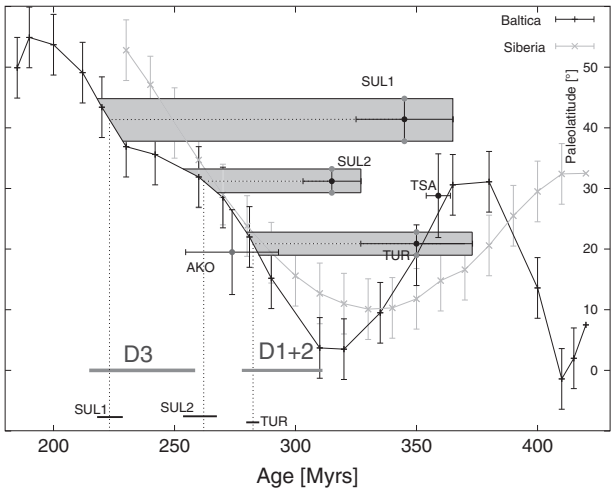


Figure 11. Plot of paleolatitudes versus age, calculated with respect to the apparent polar wander paths of Baltica [*Smethurst et al.*, 1998] and Siberia (B solution, according to *Cocks and Torsvik* [2007]) for a reference site in the Karatau mountains with a mean error of 5°. Also plotted are paleolatitudes, calculated from the regional mean directions (Table 5). Horizontal error bars define the age uncertainty. Vertical error bars are associated with a paleolatitude based on the direction at +10 and –10% untilting about the optimal untilting value (Table 5). TSA and AKO are plotted with the α_{95} confidence error. Grey boxes indicate intersection of paleolatitudes based on syn- or post-folding magnetizations from Karatau with the expected value based on the APWP of Baltica. Grey lines at the bottom (D1, D2, and D3 after *Alexeiev et al.* [2009]) mark the timing of the deformational phases in the Karatau.

temperatures within deformation zones. They show that a complex pattern can accompany orogenesis, involving different mechanism, which can lead to remagnetization, like formation of TVRM (thermo-viscous remanent magnetization) or CRM (chemical remanent magnetization) during smectite to illite conversion.

[40] It seems that in the Karatau Mountains, at least two phases of remagnetization can be identified. An early event of deformation leading to a syn-folding signal in the Bolshoi Karatau and maybe westernmost Malyi Karatau (TUR and USH). Localities SUL-1 and SUL-2, in the Malyi Karatau, were also remagnetized during a folding period. However, here the degree of unfolding, which is necessary to obtain ideal clustering of magnetic directions is lower, and the resulting syn-folding inclinations are higher. Therefore, it might reflect a different phase of remagnetization.

[41] In contrast, areas of TSA and AKO do not show any sign of remagnetization. It seems, that remagnetization events are, as proposed by *Zegers et al.* [2003] a complex superposition of different mechanisms, depending crucially on the temperature(s) of migrating fluids and deformation geometry at the specific area. Therefore, remagnetization behavior can have quite different appearance in a single orogenic.

7. Implications for the Geotectonic History

[42] In summary all paleopoles derived from Paleozoic rocks from the Karatau Range, which yield pre- or syn-folding magnetization ages, feature two characteristics in comparison with the East European apparent polar wander path [*Smethurst et al.*, 1998] as a reference:

[43] (1) A deviation in polar longitude (ΔPlong) of 25° – 65° counterclockwise with respect to the Baltic APWP present in all studied rocks (Figure 10). Thus, rotational reorganization took place at least until the early Mesozoic.

[44] (2) Analysis of the expected paleolatitudes (or inclinations) of the area in a Baltic reference frame leads to the following implications: (a) the paleolatitude of rocks from TSA are in agreement with a primary Devonian acquisition age of magnetization, (b) rocks from AKO show a slightly lower paleolatitude than expected, which can be explained by inclination shallowing, (c) rocks from TUR, since they hold a syn-folding magnetic signal, can be correlated with a remagnetization event at $\sim$ 275 – 280 Myrs ago (probably during the D1 or D2 event of *Alexeiev et al.* [2009]), (d) rocks from SUL-1 and SUL-2 areas yield a paleolatitude, which fits the expected value at $\sim$ 220–225 and $\sim$ 255–260 Myrs ago, respectively, (probably at different times during phase D3 of *Alexeiev et al.* [2009]) (Figure 11). Errors of the paleolatitudes reflect corresponding value of inclination of syn-tilt direction at ±10% of the optimal untilting percentage. This amount includes all uncertainties from the *Watson and Enkin* [1993] fold test.

[45] In this respect, we correlate the first remagnetization event with the D1 and D2 events of *Alexeiev et al.* [2009], which accompanied the initial collision of Kazakhstan and Baltica. In these phases, most of the folding amount took place. Subsequently, folding in the Chu-Sarysu basin, which took place from late-Permian to early Mesozoic times [*Abdulin et al.*, 1986; *Allen et al.*, 2001; *Alexeiev et al.*, 2009], is accompanied by remagnetization in the SUL areas. Folding in the area could have proceeded until early Mesozoic times because of a lack of unconformities in the Devonian to upper Permian sequences.

[46] The amounts of counterclockwise rotation with respect to (wrt) the APWP of Baltica decrease as a function of time of acquisition of magnetization. Devonian rocks

carrying a primary magnetization are rotated away by $\sim 65°$ (TSA), whereas rocks remagnetized during phase I of folding ($\sim$ 280 Ma) are only rotated by $\sim$ 40° and rocks magnetized during the youngest pahse (II) only show rotations in the order of $\sim 30–20°$ (Figure 10). This observation is strengthening our interpretation.

[47] The robustness of the declination deviations (ΔDec) is shown by calculating paleopoles of syn-folding directions at $\pm$20% of the optimal untilting value (Figure 10). Small differences of vertical axis rotations, between SUL1 and SUL2, and between TUR and AKO, respectively, might be explained by minor local variations of the deformational history.

[48] The amount of rotation in Karatau does not exceed $\sim 65°$, and is significantly less than maximum rotation values in the Tianshan, which can reach $\sim 90°$ [*Van der Voo et al.*, 2006]. The latter can be interpreted as evidence for an increase of maximal rotation values while following a transect from the Urals via the Karatau towards the Tianshan. Correcting for the maximal amount of rotation in the Karatau mountains suggests that the Karatau Range was originally a N-S striking fold-belt, which got subsequently bent into an "orocline" s.l. as the collision evolved through time. It might therefore represent a continuation of the Ural mountains.

8. Conclusions

[49] Primary and syn-folding magnetizations were isolated for Devonian, Carboniferous, and Permian rocks in three areas of the Karatau mountain range, based on the outcome of fold tests [*Watson and Enkin*, 1993]. The magnetizations identified in rocks of Permian and Devonian age in the Zhanatas and Akkol areas pass the fold test and are therefore interpreted to be of primary origin. The resulting mean direction after full tilt correction is $D = 210.4°$, $I = –35.3°$, $k = 74.9$, and $\alpha_{95} = 7.0°$ for the Permian rocks (AKO) and $D = 208.2°$, $I = –47.7°$, $k = 66.3$, and $\alpha_{95} = 6.9°$ for the Devonian rocks (TSA). The obtained paleolatitudes of TSA are in agreement with corresponding values of a reference site within the Karatau mountains based on the Baltica APWP [*Smethurst et al.*, 1998]. AKO yield a slightly underestimated paleolatitude compared to the APWP, which might reflect inclination shallowing.

[50] Magnetic directions of rocks from the area TUR of Bolshoi Karatau and SUL-1, SUL-2 of Malyi Karatau indicate a syn-folding age of acquisition of the magnetic signal, at $\sim$ 58% (for TUR), and $\sim$ 23%, $\sim$ 25% (for SUL-1 and SUL-2) of untilting, respectively. The syn-folding magnetic mean directions are $D = 196.9°$, $I = –37.0°$, $k = 60.4$, and $\alpha_{95} = 4.6°$ for TUR, $D = 198.5°$, $I = –60.4°$, $k = 134.5$, and $\alpha_{95} = 3.6°$ for SUL-1 and $D = 207.5°$, $I = –49.8°$, $k = 43.6$, and $\alpha_{95} = 6.4°$ for SUL-2. Because the major folding in this area started at $\sim$ 315 Myrs ago [*Alexeiev et al.*, 2009], the resulting paleolatitudes of these areas intersect the paleolatitude curve of a reference site within Karatau Mountains of the Baltica APWP at $\sim$ 280 Myrs (TUR), at $\sim$ 260 (SUL-2) and at $\sim$ 230 Myrs (SUL-1).

[51] These observations are in agreement with major phases of deformation (D1,D2, and D3 after *Alexeiev et al.* [2009]; *Rolland et al.* [2013]) in this area and the Eastern Urals-Transuralian Fault system [*Echtler and Hetzel*, 1998]. They indicate a series of distinct remagnetization events, which form a complex pattern within this orogenic environment.

[52] The whole region was, during all that time (middle Bashkirian early Mesozoic), exposed to rotational reorganization connected with the appearance of major strike slip fault systems. The Karatau mountain range shows vertical axis rotations reflected in ΔDec compared to expected values in a counterclockwise sense with amounts in a range between 20° and 60°. The amounts of observed vertical axis rotation decrease with decreasing magnetization age. The Devonian rocks from Zhanatas (TSA) show the largest amount of declination deviation ($\sim$ 60°), whereas ΔDec observed at SUL-1, SUL-2, and AKO are in the order of about 20°

[53] By comparing the new results with studies from the Tianshan and South Kazakhstan, a similar tectonic evolution is noticeable. The maximal amount of observed rotations, however, increases from the Karatau towards the Tianshan, where ΔDec reach values more than 90° [*Van der Voo et al.*, 2006]. In the Tianshan the rotational history seems, however, much more complicated than in the Karatau mountains.

[54] Our data identify the Karatau Range as the structural linkage between the Ural mountains and the CAOB basement of the Tianshan mountains relative to their Late Paleozoic to Early Mesozoic evolution. Though the Main Karatau Fault was reactivated during Neogene Tianshan orogeny, the Paleozoic sedimentary record shows the character of a major discontinuity already since Early Paleozoic. Late Carboniferous onset of contractional tectonics is shown by syn-deformational remagnetization followed by more remagnetization events during Late Permian and maybe Triassic, respectively.

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